

CSE 331: Microprocessor Interfacing and Embedded Systems

Project Report Guide

Fall 2025

Embedded System with IoT and Integrated Android App

1. Title Page

- **Project Title:** Android App Controlled Smart Car Using Raspberry pi pico.
- **Project Group/Lead/Submitter:** Ashraful Haque Lipu(2131081642),Kismat Jahan Meem(2131046642).
- **Course/Section/Department:** CSE 331: Microprocessor Interfacing and Embedded Systems / 01 / ECE.
- **Date:** 26.12.25

2. Project Abstract (Summary)

This project presents the design and implementation of a Bluetooth-controlled smart car using a microcontroller-based embedded system and an Android mobile application. The car can be controlled wirelessly through an Android app using Bluetooth communication, allowing forward, backward, left, right, and stop movements. The system integrates hardware components such as a microcontroller, motor driver, DC motors, Bluetooth module, and power supply with embedded software and a mobile application interface. The primary goal of this project is to demonstrate practical microprocessor interfacing, wireless communication, and real-time control in an embedded environment. The project has educational significance and practical relevance in robotics, automation, and IoT-based control systems.

3. Project Introduction/Objectives/Motivations/Contributions

Introduction:

The high pace of embedded systems and mobile technologies has made it possible to create intelligent and remote controlled devices. A smart car that is controlled by Bluetooth is one of the applications of this, and it is a combination of microprocessor interfacing, wireless communication, and real-time control. The project is aimed at designing and developing an embedded system which will be operated with the help of an Android smartphone application. The system combines a microcontroller, Bluetooth module, motor driver and DC motors to illustrate fundamental principles of CSE 331 such as hardware interfacing, serial communication and embedded software development. The project is a good learning tool to apply the functionality of embedded systems in relation to mobile application applications in the real world.

Objectives:

- **Objective 1:** To design a microcontroller-based smart car capable of wireless control

using Bluetooth communication.

- **Objective 2:** To develop an Android application that sends real-time control commands to the embedded system.
- **Objective 3:** To interface motors, Bluetooth module, and power management circuitry with the microcontroller.

Motivation :

This project is motivated by a desire to have practical experience in interfacing a microprocessor, wireless communication and embedded system design. The use of Bluetooth-controlled vehicles is a pillar to high-tech robotics and autonomous systems.

Contributions:

- Implementation of a low-cost Bluetooth-based control system.
- Integration of embedded hardware with an Android application.
- Practical demonstration of real-time command execution using a microcontroller.

4. Project Background

Context and Rationale:

Due to the blistering development of embedded systems and IoT, the possibility of controlling devices wirelessly has become even more significant. Control systems with Bluetooth are easy and economical to use in intricate short-range communication. The project aims to cover the necessity of applied studies in interfacing embedded systems and control (on mobile platforms).

Historical and Scientific Background:

The first robots controlled systems were based on wired communication. The advent of wireless technologies like Bluetooth permitted the ability to make the solutions of control flexible and mobile. This project is scientifically grounded on the principles of digital communication, motor control, and microcontroller programming.

Significance to the Broader Domain:

The project adds to the more general branch of embedded systems, robotics, and IoT as it has shown how mobile applications can be utilized to operate hardware devices effectively. It is the future of such developments like autonomous cars and intelligent robots.

5. Literature Review

Existing Knowledge and Theoretical Basis:

Mobile robots based on embedded systems have received extensive research in robotics, automation, and IoT. Microcontrollers and single-board microcontrollers are typically used to drive DC motors by using motor driver circuits like L293D and L298N. Bluetooth technology and especially modules such as HC-05 and HC-06 have been widely employed in short-range wireless communication because it consumes low power, is simple and very inexpensive. The serial communication via UART will serve as the theoretical basis of allowing the exchange of data between Bluetooth and the microcontroller. Android applications are commonly deployed to the software side, where they are typically used as human-machine interfaces (HMI) applications to transmit control commands wirelessly. Robotic motion control logic is usually grounded in digital signal processing, logic control as well as Taylor-principles of control systems.

Gaps and Limitations in Previous Studies:

Even though there are numerous projects of Bluetooth-controlled robotic cars, most of the previous activities are based on a limited functionality without structural design approach and without testing approaches and documentation. Some projects do not have scalability, software that was designed in a modular way, and even power management features. Also, most of them are based on off-the-shelf libraries without providing a description of the conceptualization of the underlying microprocessor interfacing, which makes their educational usefulness in academia restricted. Other aspects of security, robustness and structured evaluation metrics are also not commonly considered in previous studies.

Relevant Case Studies or Applications:

Robotic vehicles controlled by Bluetooth have been used in different areas like educational robots kits, remote-inspection robots and assistive mobility aid. Their use in the teaching of embedded systems, wireless communication and real-time control has been proven to be effective in case studies. The advanced applications such as autonomous navigation, obstacle avoidance robots and IoT-enabled surveillance platforms have also been prototyped on similar systems. These uses confirm the feasibility and usefulness of robotic control systems that use Bluetooth.

Project Relativity and Advantage:

The proposed project is based on the current research and it mitigates some of the limitations that have been identified. The project is based on a systematic design and development approach in accordance with the academic standard, unlike many others in the past. Raspberry Pi Pico is better in terms of efficiency in processing and allows flexibility in the programming language either with MicroPython or Embedded C. The usability and real-time responsiveness are enhanced with the incorporation of a tailored Android application. Moreover, this project focuses on clarity of documentation, testing procedure and knowledge of interfacing with microprocessors at the system level. This leads to a project that is more educative, more reliable, and a robust base to develop further expansion of the project to include the Internet of Things control, sensor networking, and semi-autonomous operation.

6. Design Methodology/Approach

Overview of Approach:

The project follows a modular design approach where hardware and software components are developed and tested independently before system integration.

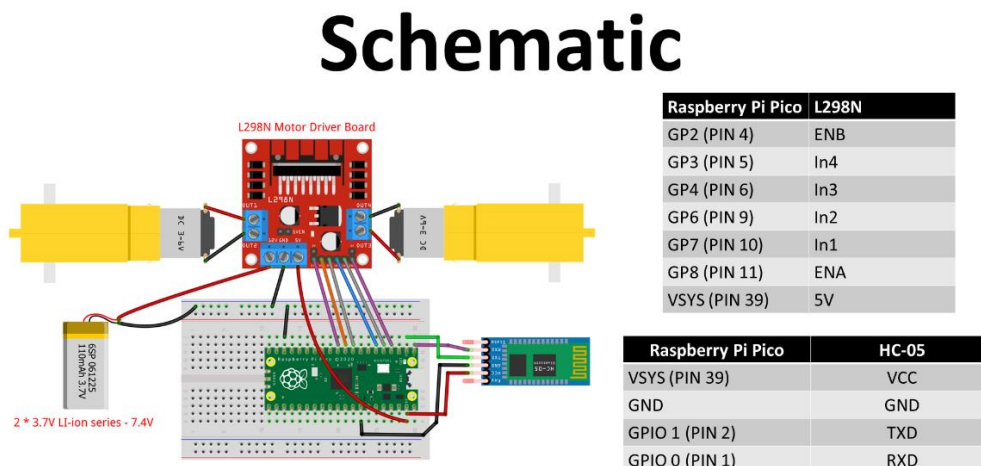
Project Phases:

- Phase 1: Planning and Design – Requirement analysis, component selection, circuit design.
- Phase 2: Hardware Implementation – Interfacing microcontroller, motor driver, Bluetooth module, and motors.
- Phase 3: Software Development – Microcontroller programming and Android app development.
- Phase 4: Testing and Integration – System testing, debugging, and performance evaluation.

Tools and Resources:

- Raspberry Pi Pico microcontroller with header pins and USB cable
- L298N motor driver controller board
- Two (2) 18650 battery storage box cases
- DC electric motors with plastic TT motor tire wheels
- Male–Female and Male–Male jumper wires
- Single wheel light-duty swivel plate (caster wheel)
- Nuts and bolts assortment kit
- Electrical cable connectors (CH2 type)
- 1 × 2-meter 20 AWG single-core wire.

Schematic:



7. Testing Methodology

How Success Will Be Measured:

- Accurate execution of movement commands from the Android app.
- Reliable Bluetooth connectivity within the specified range.
- Achievement of all defined project objectives.

Monitoring and Reporting:

Project monitoring is carried out continuously throughout all phases to ensure that the project progresses according to the planned schedule and objectives. Weekly progress reviews are conducted to evaluate task completion, identify technical issues, and verify alignment with project goals. Each phase concludes with a milestone review where deliverables such as circuit design, hardware integration, software functionality, and testing results are assessed.

Progress reporting is maintained through structured documentation, including development logs, test reports, and version updates of both hardware and software components. Any deviations from the planned timeline or unexpected technical challenges are recorded and addressed through corrective actions. Regular demonstrations of partial system functionality (such as motor control and Bluetooth connectivity) are used to validate incremental progress. This systematic monitoring and reporting approach ensures transparency, accountability, and timely completion of the project while maintaining quality and reliability.

8. Project Timeline

- **Phase 1:** [20.11.25 – 27.11.25] (Planning and Design).
- **Phase 2:** [28.11.25 – 03.12.25] (Hardware Implementation).
- **Phase 3:** [04.12.25 – 11.12.25] (Software Development)
- **Phase 4:** [12.12.25 – 20.12.25] (Testing and Integration)
- **Final Deliverable:** [24.12.25]

9. Budget

- **Personnel:** BDT 0 (Student Project)
- **Materials/Equipment:** BDT 2,200 (Motors, Bluetooth module, chassis, battery)
- **Software/Tools:** BDT 0 (Free/Open-source tools)
- **Miscellaneous:** BDT 300
- **Total Budget:** BDT 2,500

10. Team Members and Roles

- **Project Manager:** Ashraful Haque Lipu – Overall project coordination and documentation.
- **Embedded Developer:** Ashraful Haque Lipu– Hardware interfacing and microcontroller programming.
- **App Developer:** Kismat Jahan Meem – Android app design and Bluetooth communication.

11. Risks and Mitigation Strategies

- **Risk 1:** Bluetooth connection instability
Mitigation Strategy: Use proper baud rate configuration and error handling.
- **Risk 2:** Motor driver overheating
Mitigation Strategy: Ensure proper power ratings and heat dissipation.

12. Conclusion and Limitations

In this project, the authors were able to demonstrate a smart car that is controlled by Bluetooth, and it is based on the embedded system and Android application. It lays emphasis on real world microprocessor interfaces, wireless communication and integration of the system. Although the project is restricted to manual control and short-range communication, it serves as a solid ground of future development, including the use of IoT control, integration of sensors, and autonomous navigation. The project can be developed and elaborated to have conference or journal publication.

13. References

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